MOMENT OF INERTIA (I)

This is a measure of an object's resistance to rotational motion about a given axis

It depends on the mass of the object and the relative mass distribution about the axis of rotation.

$$I = \sum m_i$$

Or

$$I = \int r^2 \, dm$$

For a continuous mass distribution, where m_i is the mass of a small element, r_i is the perpendicular distance from the axis of rotation to the element and dm is an infinitesimal mass element in the continuous case.

Example of moment of inertia are:

I = $\frac{1}{2}ML^2$ ---------- for slender rod axis through center

I = $\frac{1}{2}ML^2$ ---------- for slender rod axis through one end

I = $\frac{1}{2}M(a^2 + b^2)$ ---------- for rectangular plate axis through center

I = $\frac{1}{3}Ma^2$ ---------- for thin rectangular plate axis along edge

I = $\frac{1}{2}M({R_1}^2 + {R_2}^2)$ ---------- for hollow cylinder

I = $\frac{1}{2}MR^2$---------- for solid cylinder

I = MR^2 ---------- for thin-walled hollow cylinder

I = $\frac{2}{5}MR^2$---------- for solid sphere

I = $\frac{2}{3}MR^2$---------- for thin-walled hollow sphere

Examples

1. A baton made up of four masses fastened to the end of four (4) light rods of 1 m long each. Find the moment of inertia about the system about an axis perpendicular and through the point where the rods cross (take m_i = 0.2 kg, m_2 = 0.3 kg, m_3 = 0.2 kg and m_4 = 0.3 kg)

Solution

$$I = \sum m_i\, r^2 = m_1 r_1^2 + m_2\, r_2^2 + m_3\, r_3^2 + m_4 r_4^2$$

$$= (0.2\text{kg})(0.5\ m)^2 + (0.3\text{kg})\ (0.5\ m)^2 + (0.2\text{kg})\ (0.5\ m)^2 + (0.3\text{kg})\ (0.5\ m)^2$$

$$= 0.15\ \text{kg}m^2$$

2. If m_1 = 0.2 kg m_2 = 0.3 kg

r_1 = 0 r_2 = 0.5 m

m_3 = 0.2 kg m_4 = 0.3 kg

r_3 = 0 r_4 = 0.5 m

Calculate the moment of Inertia

Solution

$$I = \sum m_i\, r^2 = m_1 r_1^2 + m_2\, r_2^2 + m_3\, r_3^2 + m_4 r_4^2$$

$$= (0.2\text{kg})(0)^2 + (0.3\ \text{kg})(0.5)^2 + (0.2\ \text{kg})(0)^2 + (0.3)(0.5\ m)^2$$

$$= 0.0056\ \text{kg}m^2$$

Practice Questions

1. What is the moment of inertia of a platform of mass M = 100 kg and radius R = 2m
2. A student of mass m = 60 kg walks slowly from the rim of a disk toward the center with a radius R = 2m. Calculate his moment of inertia.

GYROSCOPE

A gyroscope is a device for measuring or maintaining orientation based on the principles of angular momentum.

Mechanically, a gyroscope is a spinning wheel or disk in which the axle is free to assume any orientation.

Although, this orientation does not remain fixed, it changes in response to an external torque much less and in different direction than it would without the large angular momentum associated with the disk’s high rate of spin and moment of inertia.

Example

A gyroscope spins with its tip on the ground and is spinning with negligible frictional resistance. The disk of the gyroscope has 0.3 kg and is spinning at 20 rev/s. Its center of mass is 5.0 cm from the pivot and the radius of the disc is 5.0 cm. What is the precessional period of the gyroscope?

Solution

The moment of inertia of the disk is

$$\mathrm{I} = \frac{1}{2}mr^2 = \frac{1}{2}(0.30\ kg)(0.05m)^2$$

I = 3.75 x $10^{-4}\ kgm^2$

- The angular velocity of the disk is

$$20.0 \text{ rev/s} = (20.0)(2\pi) \text{ rad/s} = 125.66 \text{ rad/s}$$

The precessional angular velocity is

$$w_p = \frac{rM_g}{I_w} = \frac{(0.05m)(0.3kg)(9.8\, ms^{-2})}{(3.75x10^{-4}\, kgm^2)(125.66\, rads^{-1})}$$

$$= 3.12 \text{ rad/s}$$

The precessional period of the gyroscope is

$$T_p = \frac{2\pi}{3.12\, rad/s}$$

$$= 2.0\text{s}$$

Question

A tip has a precession frequency of 5.0 rad/s on Earth. What is its precession frequency on the moon?

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